

LEXAN* 920 Resin

Wednesday, April 27, 2011

General Information

Product Description

Opaque colors, low viscosity, superior flame retardance.

General

Material Status	• Commercial: Active
Availability	• North America
Additive	• UV Stabilizer
Features	• Flame Retardant • Low Viscosity
Appearance	• Colors Available • Opaque
Processing Method	• Injection Molding
Multi-Point Data	• Coefficient of Thermal Expansion vs. Temperature (ASTM E831) • Flexural DMA (ASTM D4065) • Instrumented Impact (Energy) (ASTM D3763) • Instrumented Impact (Load) (ASTM D3763) • Pressure-Volume-Temperature (PVT - Zoller Method) • Shear DMA (ASTM D4065) • Specific Heat vs. Temperature (ASTM D3417) • Tensile Fatigue • Tensile Stress vs. Strain (ASTM D638) • Thermal Conductivity vs. Temperature (ASTM E1530) • Viscosity vs. Shear Rate (ASTM D3835)

ASTM & ISO Properties ¹

Physical	Nominal Value	Unit	Test Method
Specific Gravity	1.21 1.22	g/cm ³	ASTM D792
Specific Volume	0.835	cm ³ /g	ASTM D792
Melt Mass-Flow Rate (MFR) (300°C/1.2 kg)	15	g/10 min	ASTM D1238
Molding Shrinkage - Flow (3.20 mm)	0.50 to 0.70	%	Internal Method
Water Absorption (24 hr)	0.15	%	ASTM D570
Water Absorption			ASTM D570
Equilibrium, 23°C	0.35	%	
Equilibrium, 100°C	0.58	%	
Mechanical	Nominal Value	Unit	Test Method
Tensile Strength ² (Yield)	62.1	MPa	ASTM D638
Tensile Strength ² (Break)	55.8	MPa	ASTM D638
Tensile Elongation ² (Yield)	7.0	%	ASTM D638
Tensile Elongation ² (Break)	90	%	ASTM D638
Flexural Modulus ³ (50.0 mm Span)	2240	MPa	ASTM D790
Flexural Strength ³ (Yield, 50.0 mm Span)	91.0	MPa	ASTM D790

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SABIC Innovative Plastics - Polycarbonate

Mechanical	Nominal Value	Unit	Test Method
Taber Abrasion Resistance 1000 Cycles, 1000 g, CS-17 Wheel	10.0	mg	ASTM D1044
Impact	Nominal Value	Unit	Test Method
Notched Izod Impact (23°C)	640	J/m	ASTM D256
Unnotched Izod Impact (23°C)	3200	J/m	ASTM D4812
Gardner Impact (23°C)	169	J	ASTM D3029
Tensile Impact Strength ⁴	525	kJ/m ²	ASTM D1822
Hardness	Nominal Value	Unit	Test Method
Rockwell Hardness M-Scale R-Scale	70 118		ASTM D785
Thermal	Nominal Value	Unit	Test Method
Deflection Temperature Under Load 0.45 MPa, Unannealed, 6.40 mm	138	°C	ASTM D648
Deflection Temperature Under Load 1.8 MPa, Unannealed, 6.40 mm	132	°C	ASTM D648
Vicat Softening Temperature	152	°C	ASTM D1525 ⁵
CLTE - Flow (-40 to 95°C)	0.000068	cm/cm/°C	ASTM E831
Thermal Conductivity	0.19	W/m/K	ASTM C177
Electrical	Nominal Value	Unit	Test Method
Volume Resistivity	> 1.0E+17	ohm·cm	ASTM D257
Dielectric Strength (3.20 mm, in Air)	17	kV/mm	ASTM D149
Dielectric Constant 50 Hz 60 Hz 1 MHz	3.01 3.01 2.96		ASTM D150
Dissipation Factor 50 Hz 60 Hz 1 MHz	0.00090 0.00090 0.010		ASTM D150
Arc Resistance (PLC) ⁶	PLC 7		ASTM D495
Flammability	Nominal Value	Unit	Test Method
Flame Rating - UL (1.47 mm)	V-0		UL 94
Oxygen Index	35	%	ASTM D2863
UL	Nominal Value	Unit	Test Method
RTI Str	130	°C	UL 746
RTI Imp	120	°C	UL 746
RTI Elec	130	°C	UL 746
Comparative Tracking Index (CTI) (PLC)	PLC 2		UL 746
High Voltage Arc Tracking Rate (HVTR) (PLC)	PLC 4		UL 746
Hot-wire Ignition (HWI) (PLC)	PLC 2		UL 746
High Amp Arc Ignition (HAI) (PLC)	PLC 3		UL 746

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SABIC Innovative Plastics - Polycarbonate

UL	Nominal Value	Unit	Test Method
Outdoor Suitability	f1		UL 746C

Processing Information

Injection	Nominal Value	Unit
Drying Temperature	121	°C
Drying Time	3.0 to 4.0	hr
Drying Time, Maximum	48	hr
Suggested Max Moisture	0.020	%
Suggested Shot Size	40 to 60	%
Rear Temperature	266 to 288	°C
Middle Temperature	277 to 299	°C
Front Temperature	288 to 310	°C
Nozzle Temperature	282 to 304	°C
Processing (Melt) Temp	288 to 310	°C
Mold Temperature	71.1 to 93.3	°C
Back Pressure	0.345 to 0.689	MPa
Screw Speed	40 to 70	rpm
Vent Depth	0.025 to 0.076	mm

Notes

¹ Typical properties: these are not to be construed as specifications.

² Type I, 50 mm/min

³ 1.3 mm/min

⁴ Type S

⁵ Rate B (120°C/h), Loading 2 (50 N)

⁶ Tungsten Electrode

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